TDT4258 Lab 3

Embedded Device Programming

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Agenda

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3. Sense Hat and its components
4. Main task: Tetris on the Sense Hat
5. Framebuffer basics
6. Joystick input handling
7. Colourising the LED matrix
8. Recommended approach
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Lab 3 Overview

- **Goal:** Program an **embedded device** and interact with hardware
- **Hardware:** Raspberry Pi 4 + Sense Hat
- **Main task:**
  - Show Tetris game on the **LED matrix**
  - Control the game using the **Sense Hat joystick**
- **Language:** C (modify provided `stetris.c`)
- **Grading:** 10 points for a fully working solution
- **Plagiarism:** Discuss ideas ✅, share code ❌

Setting up the Raspberry Pi 4

- Provided hardware:
  - Raspberry Pi 4, Sense Hat, SD card, power supply
- First step: **Flash Raspberry Pi OS (32-bit)** onto the SD card
  - Use Raspberry Pi Imager
- In Imager:
  - Set hostname
  - Enable SSH
  - Configure Wi-Fi (⚠️ No Eduroam)
- Boot the Pi:
  - Rainbow LED → then white square (Sense Hat ready)

⚠️ **SD card orientation:**

Insert the card with the **label side facing down** toward the Pi board.

Forcing it in the wrong way can **damage the card slot**.

Working on the Raspberry Pi

Two main options:

1. **Desktop mode:**
   Connect monitor, keyboard, mouse → GUI interface
2. **Remote via SSH:**
   - Connect over local Wi-Fi/Ethernet
   - Find Pi's IP or use the hostname you set
   - Use terminal or SSH client (Linux/Mac: built-in; Windows: WSL or PuTTY)

💡 Make sure the Pi has internet access for updates and packages.

Sense Hat Overview

- **8×8 RGB LED Matrix**
  - 16-bit **RGB565** colors: 5 bits red, 6 bits green, 5 bits blue
- **5-way Joystick**
  - Up, down, left, right → arrow keys
  - Press → Enter key

Main Task

Implement **Tetris** on the Sense Hat:

- Display the **playfield** on the LED matrix
  - Only show the tiles, no score or stats
- Control game using joystick:
  - Left/Right → move tile
  - Down → drop tile
  - Up → pause game
  - Up again → unpause game
  - Center press → exit game
- Tiles must move **continuously** if joystick is held down
- **Keep each tile's color fixed** until it disappears

Framebuffer Basics

- The LED matrix is exposed as a **Linux framebuffer**:
  - Device identified as `RPi-Sense FB`
- Use:
  - `mmap()` / `munmap()` → map device memory
  - `ioctl()` → query device settings
- ⚠️ Do **not** hard-code `/dev/fb0` → check device identification

Joystick Input

- Joystick is exposed as a **Linux input event device**
  - Identified as `Raspberry Pi Sense HAT Joystick`
- Use:
  - `/dev/input/event*` interface
  - `poll()` to check for new events (non-blocking)
- ⚠️ Don’t assume fixed path; find device by name at runtime
- Must handle continuous input smoothly → no pauses in gameplay

Colourising the Playfield

- Each tile should have a **random or varying color**
- Color should **not change** while tile is on the board
- Ensure colors are **easy to distinguish**
- Example approach:
  - Pick color from a predefined palette or generate via RNG

Recommended Approach

1. Verify the provided **Tetris code** works on console
2. Implement **Sense Hat LED rendering** (framebuffer)
3. Implement **joystick input**
4. Add **color assignment** for tiles
5. Test step-by-step on the Pi
6. Keep console output **unchanged**

Optional Tasks

- **A:** Light entire LED matrix with one color, change color with joystick press
- **B:** mini-Pac-Man
- **E:** Extend Tetris with **multiple shapes**

Submission Requirements

- Submit:
  - `stetris.c` (commented & tidy)
  - **AI-statement**
- Program must:
  - Still show console output
  - Preserve original game logic & timing
- **Deadline:** Friday, 23rd October 17:00
- Late or broken submissions → reduced points

Common Pitfalls

- Hard-coding framebuffer or input paths
- Blocking joystick reads → freezes gameplay
- Forgetting to continuously move tile on held joystick
- Using colors that are too similar
- Forgetting to keep console output intact
- Lack of comments → harder to award points